HAO LUO

Ph.D. Candidate in Mechanical Engineering · Tsinghua University

Living DNA Data Storage · Engineered Living Materials · Microfluidics · Biofabrication · Automation

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PROFILE

Ph.D. candidate developing physically addressable and regenerative living systems for DNA data storage. Research spans digital-to-DNA encoding, plasmid and bacterial engineering, droplet microfluidics, living microspheroid fabrication, fluorescence-assisted random access, dry-state preservation, information regeneration and rewriting, and automated system integration. First author of two *Advanced Materials* papers introducing the ELMM archival file system and regenerative Living Disk-Drive architecture.

EDUCATION

Tsinghua University — 2021—2026

Ph.D. in Mechanical Engineering · Advisor: Prof. Zhuo Xiong · GPA: 3.8 / 4.0

- Expected graduation: December 2026 · Beijing Key Laboratory of Intelligent Organ Biofabrication and Regenerative Repair.

Tsinghua University — 2017—2021

B.Eng. in Mechanical Engineering · GPA: 3.77 / 4.0 · Rank: 3 / 106

SELECTED RESEARCH CONTRIBUTIONS

Regenerative Living Disk-Drive for DNA Data Storage — 2024—2025

System architecture, process development, and instrument integration

- Connected a lyophilized Living Disk, fluorescence-based Optical Retriever, and desktop Living Drive for release, bacterial expansion, re-encapsulation, and database replenishment.
- Validated 10 automated regeneration cycles, 13 lyophilization-rehydration cycles, and four months of ambient dry storage; established a CRISPR-Cas12a/λ-Red rewriting interface.

Engineered Living Memory Microspheroid (ELMM) Archival File System — 2023—2024

Full-stack process and algorithm development

- Developed DNA encoding/decoding, functional plasmid indexing, droplet-microfluidic fabrication of ~50 µm living file units, Boolean retrieval, and freeze-drying workflows.
- Demonstrated >98% fluorescence-assisted sorting accuracy and recoverable information after seven lyophilization-rehydration cycles.

Patient-derived Cancer Assembloids — 2021—2022

Microfluidic platform and fabrication process development

- Developed droplet-microfluidic processes for patient-derived lung and gastric cancer assembloids used in tumor-microenvironment modeling and drug screening.

Chip-based High-throughput DNA Synthesizer — 2022—2023

Automation prototype development

- Developed a chip-assisted synthesis prototype and collaborative encoding method; validated 13-nt DNA synthesis.

PEER-REVIEWED PUBLICATIONS

[1] **Hao Luo**, JinKai Gao, XiangXiang Huang, YongCong Fang, TianYu Huang, YingKai Xia, ZeYang Yu, ChengHao Cao, and Zhuo Xiong. **Thermo-Responsive Living Microspheroids Enable a Regenerative Living Disk-Drive System for DNA Data Storage.** *Advanced Materials* 38(42), e73806 (2026). doi:10.1002/adma.73806. **FIRST AUTHOR**

[2] **Hao Luo**, Wen Huang, ZhongHui He, Yongcong Fang, Yueming Tian, and Zhuo Xiong. **Engineered Living Memory Microspheroid-Based Archival File System for Random Accessible In Vivo DNA Storage.** *Advanced Materials* 37(13), 2415358 (2025). doi:10.1002/adma.202415358. **FIRST AUTHOR**

[3] Yanmei Zhang, Qifan Hu, Yuquan Pei, **Hao Luo**, et al. **A Patient-Specific Lung Cancer Assembloid Model with Heterogeneous Tumor Microenvironments.** *Nature Communications* 15, 3382 (2024). doi:10.1038/s41467-024-47737-z.

[4] Xinxin Xu, Yunhe Gao, Jianli Dai, Qianqian Wang, Zixuan Wang, Wenquan Liang, Qing Zhang, Wenbo Ma, Zibo Liu, **Hao Luo**, et al. **Gastric Cancer Assembloids Derived from Patient-Derived Xenografts: A Preclinical Model for Therapeutic Drug Screening.** *Small Methods* 8(9), 2400204 (2024). doi:10.1002/smtd.202400204.

[5] Min Ye, Yiran Shan, Bingchuan Lu, **Hao Luo**, et al. **Creating a Semi-Opened Micro-Cavity Ovary Through Sacrificial Microspheres as an In Vitro Model for Discovering the Potential Effect of Ovarian Toxic Agents.** *Bioactive Materials* 26, 216-230 (2023). doi:10.1016/j.bioactmat.2023.02.029.

[6] Yanmei Zhang, Zixuan Wang, Qifan Hu, **Hao Luo**, et al. **3D Bioprinted GelMA-Nanoclay Hydrogels Induce Colorectal Cancer Stem Cells Through Activating Wnt/β-Catenin Signaling.** *Small* 18(18), 2200364 (2022). doi:10.1002/smll.202200364.

CHINESE INVENTION PATENTS

[P1] Zhuo Xiong, Yanmei Zhang, Ting Zhang, **Hao Luo** (student first inventor). **Method and Device for Constructing Personalized Tumor Assembloids Based on Droplet Microfluidics.** CN113583960B / ZL202110613983.6. Granted, 2023.

[P2] Zhuo Xiong, **Hao Luo** (student first inventor), Liliang Ouyang, Yu Yang. **DNA-based Information Storage and Reading Method and Device.** CN117789788A. Published, 2024.

[P3] Zhuo Xiong, Ben Pei, Liliang Ouyang, **Hao Luo**. **Hierarchical DNA Data Storage and Reading Method.** CN115421669B / ZL202211217725.7. Granted, 2025.

[P4] Min Ye, Ting Zhang, Zhuo Xiong, Bingchuan Lu, **Hao Luo**. **Ovulating Artificial Ovary Scaffold and Its Preparation and Application.** CN114748690B / ZL202210248677.1. Granted, 2022.

[P5] Zhuo Xiong, **Hao Luo** (student first inventor). **Rewritable Engineered Living DNA Information Storage Unit and Construction Method.** Chinese invention patent application, 2026.

[P6] Zhuo Xiong, **Hao Luo** (student first inventor), XiangXiang Huang. **Hierarchical Living DNA Data Storage System and Operating Method.** Chinese invention patent application, 2026.

SELECTED ENGINEERING PROJECTS

Machine Vision Dot Recognition — 2025

Project lead · Python/OpenCV algorithm, DLL packaging, and integration

- Developed image preprocessing, feature extraction, and dot-pattern recognition; deployed the packaged module within an information system.

Laser Galvanometer Dynamic Focusing and Embedded Servo Control — 2019—2021

Core developer · National Undergraduate Innovation Project

- Integrated mechanical design, MATLAB simulation, STM32 control, sensors, actuators, part manufacturing, and prototype testing; project rated Excellent.

SELECTED TALKS & CONFERENCES

BDMC 2026 · The Chinese University of Hong Kong Jul 2026

Thermo-Responsive Living Microspheroids Enable a Regenerative Living Disk-Drive System for DNA Data Storage · Oral presentation

ICCES 2025 · Changsha, China May 2025

Engineered Living Memory Microspheroid-Based Archival File System for Random Accessible In Vivo DNA Storage · Oral presentation

ACBD-ISBM 2023 · Beijing, China Mar 2023

Research on the Fabrication of Cancer Assembloids Based on Droplet Microfluidics · Oral presentation

TEACHING

Biomaterials Engineering and Devices 2021—2026

Teaching Assistant · Tsinghua University

- Outstanding Teaching Assistant, 2022 · Evaluation in the top 5% university-wide; contributed to AI-enabled course reform and the Biomanufacturing Brain teaching platform.

Physics Laboratory B 2025—2026

Teaching Assistant · Tsinghua University

Fundamentals of Physics 2025—2026

Teaching Assistant · Tsinghua University

SELECTED HONORS & AWARDS

Outstanding Teaching Assistant	Tsinghua University · Top 5%	2022
Comprehensive Excellence Scholarship	Tsinghua University	2022
Excellent Completion · National Undergraduate Innovation Project	Laser galvanometer control system	2021
Energy Science Scholarship	Tsinghua University	2020
National Encouragement Scholarship	Two-time recipient	2018, 2019

TECHNICAL EXPERTISE

BIOENGINEERING	ENGINEERING	COMPUTING
Plasmid engineering · Bacterial culture · Droplet microfluidics · Hydrogel microspheres · Freeze-drying · Organoids / assembloids	SolidWorks · AutoCAD · ANSYS · STM32 · Arduino · CNC · 3D printing · Mechanical design · System integration	Python · OpenCV · MATLAB · C/C++ · DNA encoding · Image processing · DLL packaging · Scientific automation

ADDITIONAL EXPERIENCE

Guangdong Shunde Graduate School of Innovation 2023

R&D; Assistant · GelMA-PVA artificial vessel rapid-forming process

Silver Basis Technology 2020

Structural Design Intern · Axial cam-compression rotary engine concept and analysis

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